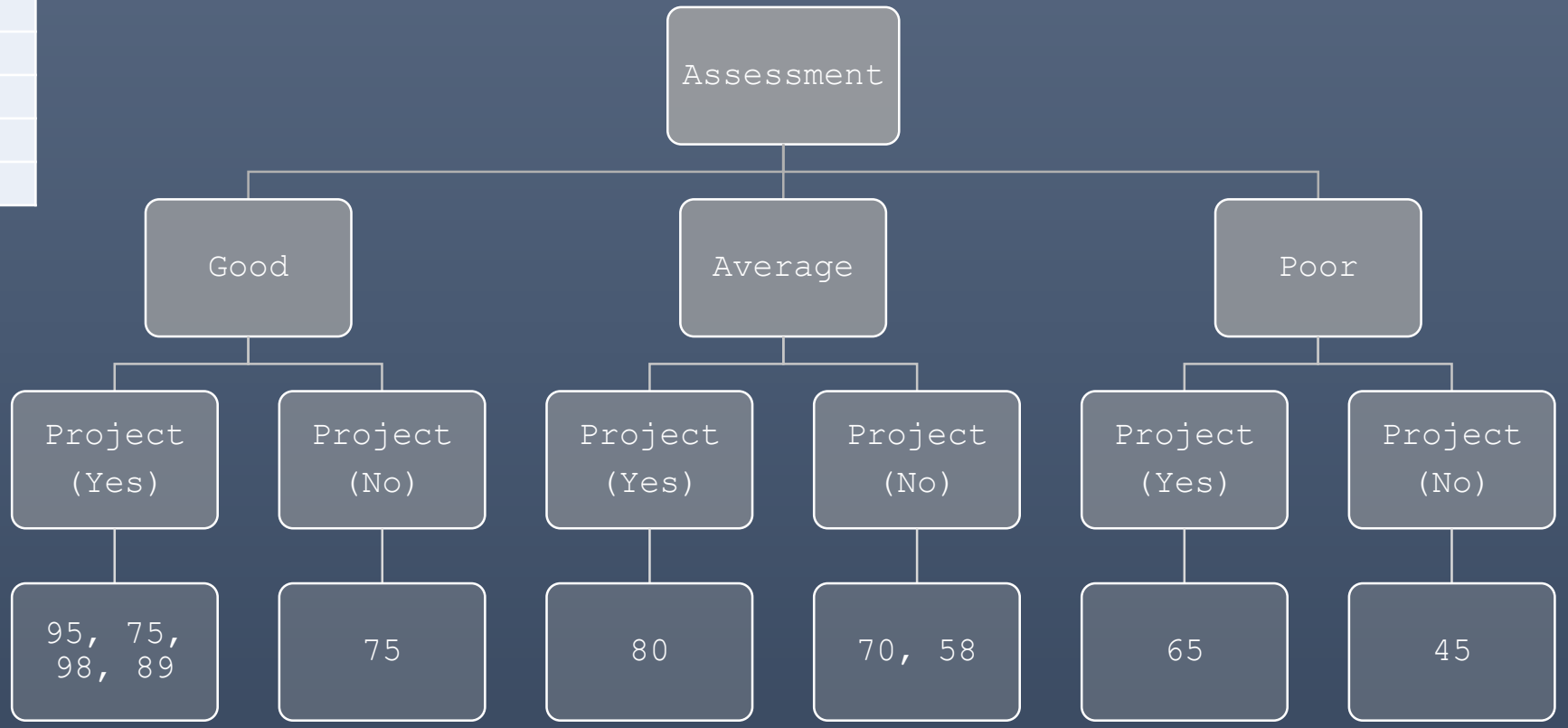
Decision Tree Regression is used to solve regression problems.
A **regression problem** means we want to **predict a numerical value**.

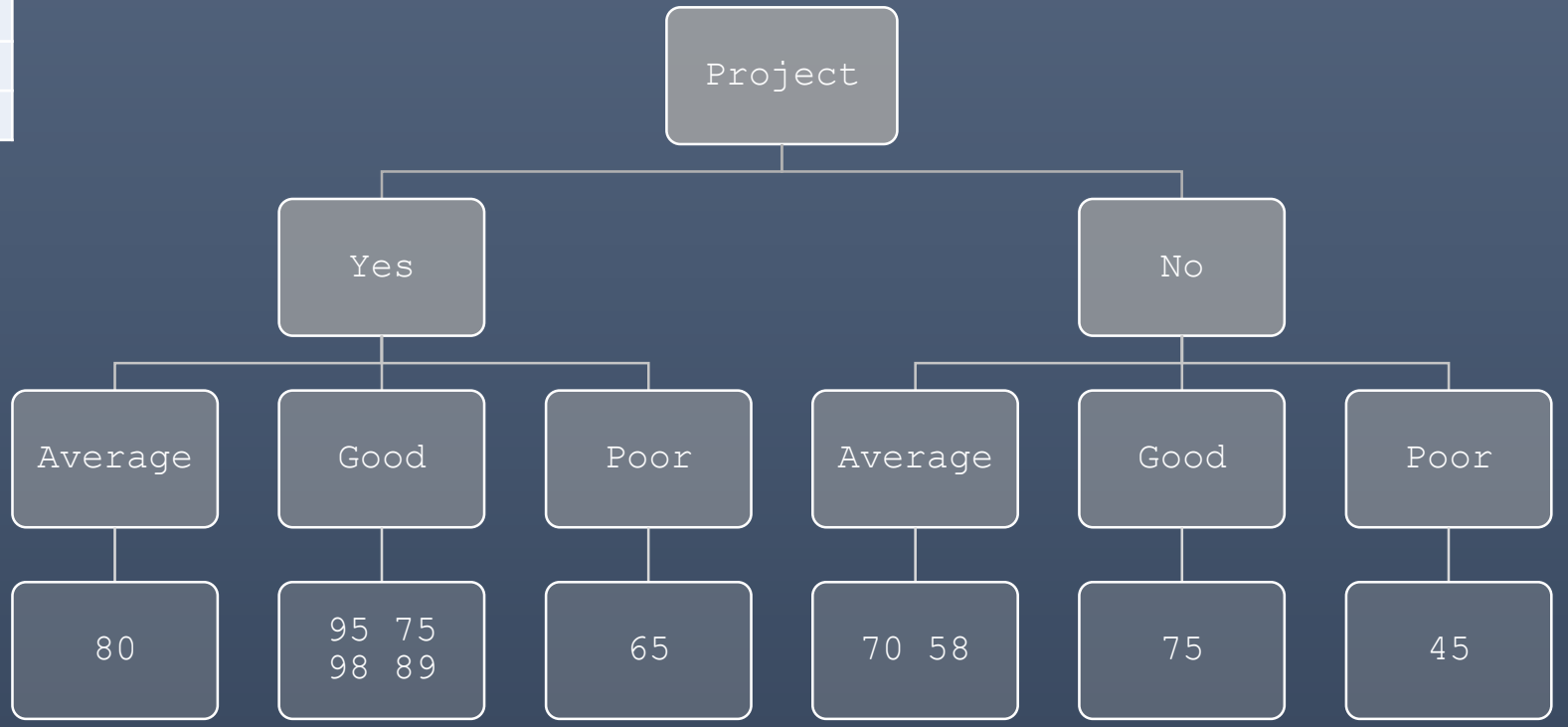
#	Assessment	Project	Result
1	Good	Yes	95
2	Average	No	70
3	Good	Yes	75
4	Poor	No	45
5	Good	Yes	98
6	Average	Yes	80
7	Good	No	75
8	Poor	Yes	65
9	Average	No	58
10	Good	Yes	89

In Decision Tree Regression (DTR), we compute the standard deviation reduction (SDR) for all features, and the feature with the highest value becomes the root node, followed by the rest.

#	Assessment	Project	Result
6	Average	Yes	80
2	Average	No	70
9	Average	No	58
1	Good	Yes	95
3	Good	Yes	75
5	Good	Yes	98
10	Good	Yes	89
7	Good	No	75
8	Poor	Yes	65
4	Poor	No	45



#	Project	Assessment	Result
2	No	Average	70
9	No	Average	58
7	No	Good	75
4	No	Poor	45
6	Yes	Average	80
1	Yes	Good	95
3	Yes	Good	75
5	Yes	Good	98
10	Yes	Good	89
8	Yes	Poor	65



#	Project	Assessment	Result
2	No	Average	70
9	No	Average	58
7	No	Good	75
4	No	Poor	45
6	Yes	Average	80
1	Yes	Good	95
3	Yes	Good	75
5	Yes	Good	98
10	Yes	Good	89
8	Yes	Poor	65

Decision Tree Regression — Steps

Identify the features → Assessment, Project.

Identify the target → Result.

Calculate the standard deviation of the Result data.

Now we need to caluclate the SDR of all features

Take 1st feature

1. **Try different splits** for each feature.
2. **Calculate the standard deviation** for each resulting group.
3. **Calculate SDR** for each split.

Repeat the Same for all feature.

Compare the SDR values.

Select the split with the highest SDR as the root node.

Calculate SD of Result

#	Assessment	Project	Result
1	Good	Yes	95
2	Average	No	70
3	Good	Yes	75
4	Poor	No	45
5	Good	Yes	98
6	Average	Yes	80
7	Good	No	75
8	Poor	Yes	65
9	Average	No	58
10	Good	Yes	89

$$x = \frac{95 + 70 + 75 + 45 + 98 + 80 + 75 + 65 + 58 + 89}{10}$$

$$= \frac{750}{10}$$

$$\bar{x} = 75$$

$$SD = \sqrt{\frac{(95-75)^2 + (70-75)^2 + (75-75)^2 + (45-75)^2 + (98-75)^2 + (80-75)^2 + (75-75)^2 + (65-75)^2 + (58-75)^2 + (89-75)^2}{10}}$$

Dividing by N

$$\frac{2464}{10} = 246.4$$

Taking the Square Root

$$\sqrt{246.4} \approx 15.70$$

Calculate SD of Assessment

#	Assessment	Project	Result
1	Good	Yes	95
3	Good	Yes	75
5	Good	Yes	98
7	Good	No	75
10	Good	Yes	89

A. Assessment = Good

Results:

$$95,\ 75,\ 98,\ 75,\ 89$$

Average

$$Average = \frac{95 + 75 + 98 + 75 + 89}{5}$$

$$\boxed{Average = 86.4}$$

Standard Deviation

$$SD_{Good} = \sqrt{\frac{(95 - 86.4)^2 + (75 - 86.4)^2 + (98 - 86.4)^2 + (75 - 86.4)^2 + (89 - 86.4)^2}{5}}$$

$$= \sqrt{\frac{475.2}{5}}$$

$$\boxed{SD_{Good} = 9.75}$$

Calculate SD of Average

#	Assessment	Project	Result
2	Average	No	70
6	Average	Yes	80
9	Average	No	58

B. Assessment = Average

Results:

$$70,\ 80,\ 58$$

Average

$$Average = \frac{70 + 80 + 58}{3}$$

$$\boxed{Average = 69.33}$$

Standard Deviation

$$SD_{Average} = \sqrt{\frac{(70 - 69.33)^2 + (80 - 69.33)^2 + (58 - 69.33)^2}{3}}$$

$$\boxed{SD_{Average} = 8.99}$$

Calculate SD of Poor

#	Assessment	Project	Result
4	Poor	No	45
8	Poor	Yes	65

C. Assessment = Poor

Results:

$$45,\ 65$$

Average

$$Average = \frac{45 + 65}{2} = 55$$

Standard Deviation

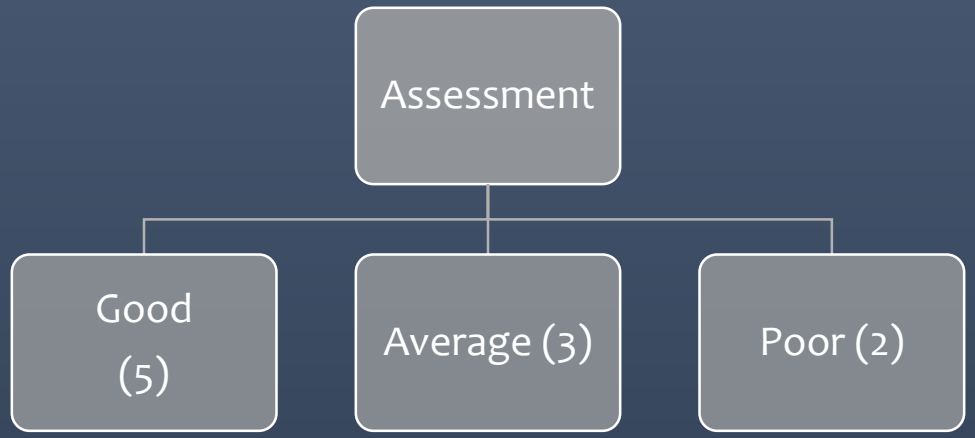
$$SD_{Poor} = \sqrt{\frac{(45 - 55)^2 + (65 - 55)^2}{2}}$$

$$= \sqrt{\frac{200}{2}}$$

$$\boxed{SD_{Poor} = 10}$$

Calculate SDR of Feature one (Assessment)

#	Assessment	Result
6	Average	80
2	Average	70
9	Average	58
1	Good	95
3	Good	75
5	Good	98
10	Good	89
7	Good	75
8	Poor	65
4	Poor	45



$$\text{Weighted SD} = \sum_{i=1}^{k} \frac{N_i}{N} \cdot SD_i$$

Where:

- N_i is the number of instances in child subset i
- N is the total number of instances in the parent dataset ($N = \sum N_i$)
- SD_i is the standard deviation of child subset i

Weighted Standard Deviation — Assessment

$$Weighted\ SD = \frac{5}{10}(9.75) + \frac{3}{10}(8.99) + \frac{2}{10}(10)$$

$$= 4.875 + 2.697 + 2$$

$$\boxed{Weighted\ SD = 9.57}$$

Standard Deviation Reduction

$$SDR = 15.70 - 9.57$$

$$\boxed{SDR_{Assessment} = 6.12}$$

4. Project

Calculate the SD of Yes

#	Assessment	Project	Result
1	Good	Yes	95
3	Good	Yes	75
5	Good	Yes	98
6	Average	Yes	80
8	Poor	Yes	65
10	Good	Yes	89

A. Project = Yes

Results:

$$95,\ 75,\ 98,\ 80,\ 65,\ 89$$

Average

$$Average = \frac{95 + 75 + 98 + 80 + 65 + 89}{6}$$

$$\boxed{Average = 83.67}$$

Standard Deviation

$$SD_{Yes} = \sqrt{\frac{(95 - 83.67)^2 + (75 - 83.67)^2 + (98 - 83.67)^2 + (80 - 83.67)^2 + (65 - 83.67)^2 +}{6}}$$

$$\boxed{SD_{Yes} = 11.54}$$

4. Project

Calculate the SD of No

#	Assessment	Project	Result
2	Average	No	70
4	Poor	No	45
7	Good	No	75
9	Average	No	58

B. Project = No

Results:

$$70,\ 45,\ 75,\ 58$$

Average

$$Average = \frac{70 + 45 + 75 + 58}{4}$$

$$\boxed{Average = 62}$$

Standard Deviation

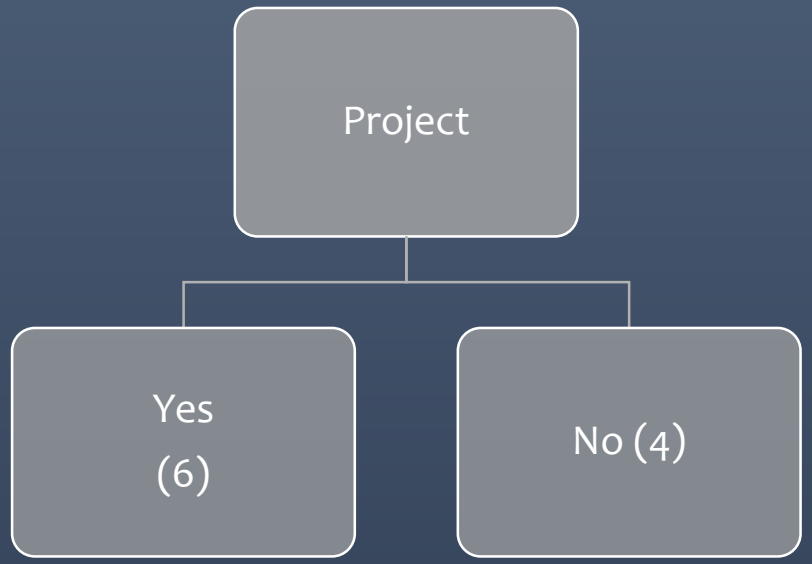
$$SD_{No} = \sqrt{\frac{(70 - 62)^2 + (45 - 62)^2 + (75 - 62)^2 + (58 - 62)^2}{4}}$$

$$= \sqrt{\frac{538}{4}}$$

$$\boxed{SD_{No} = 11.60}$$

Calculate the SDR feature-2 Project

Project	Result
Yes	80
Yes	95
Yes	75
Yes	98
Yes	89
Yes	65
No	70
No	58
No	75
No	45



Weighted Standard Deviation — Project

$$Weighted\ SD = \frac{6}{10}(11.54) + \frac{4}{10}(11.60)$$

$$= 6.924 + 4.64$$

$$\boxed{Weighted\ SD = 11.56}$$

Standard Deviation Reduction

$$SDR = 15.70 - 11.56$$

$$\boxed{SDR_{Project} = 4.13}$$

Attribute	Weighted SD	SD Reduction
Assessment	**9.57**	**6.12**
Project	11.56	4.13

Best Attribute

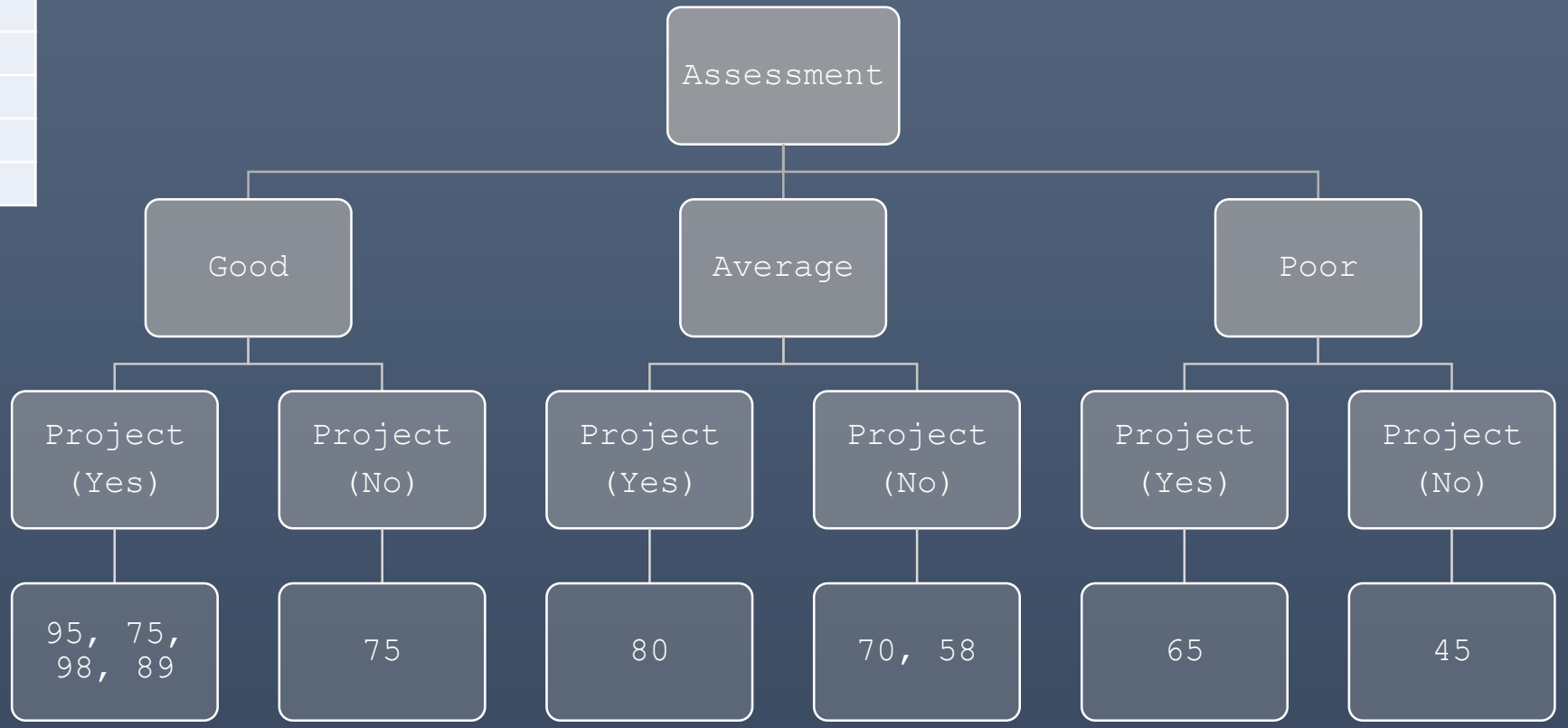
The attribute with the **highest Standard Deviation Reduction** is:

$$\boxed{\text{Assessment} = 6.12}$$

Therefore, for the **Decision Tree Regressor**, the first/root split should be:

$$\boxed{\textbf{Assessment}}$$

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1	Good	Yes	95
3	Good	Yes	75
5	Good	Yes	98
10	Good	Yes	89
7	Good	No	75
8	Poor	Yes	65
4	Poor	No	45



#	Project	Assessment	Result
2	No	Average	70
9	No	Average	58
7	No	Good	75
4	No	Poor	45
6	Yes	Average	80
1	Yes	Good	95
3	Yes	Good	75
5	Yes	Good	98
10	Yes	Good	89
8	Yes	Poor	65

